

MOHAMMAD QATAADAH ZAKIR

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer specializing in production-ready AI applications, LLM systems, retrieval-augmented generation (RAG), and scalable backend infrastructure. Experienced designing and deploying AI applications using Python, FastAPI, Docker, AWS, and vector databases, with a focus on deploying reliable, high-performance AI software solutions.

WORK EXPERIENCE

Software Engineer – Intern | Modern Arc Group

May 2024 – August 2024

- Assisted in the development and enhancement of web-based business applications using modern software engineering practices, contributing to frontend and backend development, REST API integration, SQL database management, debugging, testing, and documentation, while helping deliver 5+ client and internal software solutions within project deadlines.
- Worked alongside senior software engineers to analyze business requirements, implement new features, troubleshoot complex software defects, and participate in peer code reviews, successfully resolving 70+ reported issues and improving application performance, maintainability, and user experience across multiple development cycles.
- Contributed to software deployment, database optimization, and technical documentation while gaining experience with Git, Agile/Scrum methodologies, SDLC, version control, and cross-functional collaboration, reducing onboarding time for new developers by 30% and supporting faster feature delivery throughout the internship.

Freelance Software Engineer

June 2022 – January 2024

- Designed and deployed AI-powered applications using Python, TensorFlow/PyTorch, and Supabase (PostgreSQL, Auth, Storage), implementing LLM integrations, embeddings, vector search, and secure API endpoints, improving inference latency by ~35% and enhancing user experience.
- Built an AI-driven automation system for data extraction and workflow orchestration, leveraging LLMs, web scraping, and task scheduling (cron, queues) with REST APIs and cloud functions, reducing manual processing time by ~50% and increasing operational efficiency.
- Architected and optimized scalable AI pipelines for end-to-end model workflows, integrating data preprocessing, embedding generation, and vector database indexing (FAISS/Supabase pgvector) with FastAPI-based serving layers, improving retrieval accuracy by ~25% and system throughput by ~30% under concurrent workloads.

EDUCATION

Software Engineering, B.Eng. (Hons)

University of Greater Manchester (formerly University of Bolton) | 2025

Computer Engineering, B.Eng.

Memorial University of Newfoundland (MUN) | 2022 (Credit transfer)

PROJECTS

LLM Fine-Tuning & Inference Pipeline

Designed and deployed a production-ready large language model (LLM) training and inference pipeline, covering parameter-efficient fine-tuning, model optimization, evaluation, and high-performance deployment. The project demonstrates expertise in modern LLM engineering, scalable inference, and efficient deployment on consumer and cloud hardware. It incorporates LoRA/QLoRA fine-tuning, quantization, Hugging Face Transformers, and llama.cpp to optimize model performance while reducing computational requirements for real-world deployment.

Enterprise Retrieval-Augmented Generation (RAG) System

Built a production-grade Retrieval-Augmented Generation (RAG) platform for intelligent document search and question answering, integrating modern retrieval techniques, LLM inference, and scalable backend infrastructure. The project emphasizes low-latency retrieval, high-quality contextual responses, and cloud-ready deployment. Leveraging vector databases, semantic embeddings, and FastAPI, the system delivers context-aware responses across large document collections while minimizing hallucinations and maintaining fast query performance.

QEMI - AI Development Platform

Developed an AI-powered software engineering platform that combines real-time system monitoring, runtime diagnostics, and large language models to assist developers with intelligent debugging and code analysis. The platform was designed as a production-ready MVP and demonstrated to potential clients for early-stage commercial adoption. Built with a modern React frontend and Python-based backend, QEMI integrates AI-assisted diagnostics, automated error analysis, and developer tooling to streamline debugging workflows and improve software productivity.

SKILLS

Programming languages – Python, SQL, Typescript, Bash,

Frameworks & Libraries – PyTorch, Tensorflow, Scikit-learn, XGBoost, Hugging Face, llama-cpp, ONNX

LLMs & AI Frameworks – OpenAI API, Claude, Cursor, Embedding APIs, Vector Databases, LangChain, Together AI, Zapier, LangGraph

Cloud, Data & Infrastructure – AWS (SageMaker, EC2, S3), Azure, Docker, Kubernetes, PostgreSQL, MongoDB, Supabase, Git/Github, REST

Soft skills – Agile, Cross-functional team, Communication, Client Collaboration, Documentation, Adaptability, Time Management, Emotional Intelligence

CERTIFICATIONS

- EDUCBA – AI with Python: Foundations to Projects Specialization
- DeepLearnin.AI – Deep Learning Specialization
- Duke University – MLOps | Machine Learning Operations Specialization
- Stanford Online – Machine Learning Specialization
- HarvardX - Computer Science for Python Programming
- HarvardX - Computer Science for Artificial Intelligence
- IBM – AI Chatbots without Programming
- IBM – Python basics for Data Science
- IBM – AI Applications with Watson
- University of Minnesota – Software Development Lifecycle